MIT 18.06 SC - 1.9

Problem 8.1

a) x_p must not be multiplied by (coefficient must be 1)

b) If x_p is a particular solution then $x_p + x_n$ (where x_n is a vector in the nullspace of A) is also a particular solution.

c) Every matrix has the 0 vector in its nullspace.

Problem 8.2

$$[U\ 0] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 0 & 4 & 0 \end{array}\right]$$

$$\downarrow$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

$$\downarrow$$

rref:- $$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

Solution to $Rx = 0$ is $x = c\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$

where c is a constant $c \in R$

$$[U\ c] = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 0 & 4 & 8 \end{array}\right]$$

$$\downarrow$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$$\downarrow$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$$x_p = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

$$x_{complete} = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} + c \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

where $x_{complete}$ is the set of solutions to $Rx = d$.

We check our work by plugging our solutions into the equation $Ux = 0$ and $Ux = c$:-

$Ux = 0$:-

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$Ux = c$:-

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ 8 \end{bmatrix}$$

Problem 8.3

Yes, $A = B$, because $Ax = Cx$ for every vector x (assuming x is of the right dimensions).